Crossed wave packets in local dS_3: a spectral-cutoff witness and physical overlaps

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Working research note — 31 October 2026

Abstract

We evaluate the crossed overlap of two regulated creation sequences for a specified initial observer wave packet. Internal observer masses retain their infinite physical spectrum; finite integration boxes have analytic positive-norm tail bounds. Independently, the spherical component of the local vertex becomes an odd convolution kernel. Removing masses below the observer threshold gives an exact boundary term of definite sign. A finite-norm group-averaged test state proves a nonzero zero-time order difference. Quadrature and floating-point errors remain empirical. The setting is $G_N = 0$, not a finite-gravity Lyapunov calculation.

1 State, regulated operators, and observable

Use radius one, $G_N = 0$, proper $SO^+(3,1)$, scalar BD observers with labels at least $s = 0.2$, and distinct free KG matter species $p = 0.6$, $q = 0.9$. Local couplings are one. The proper mass and vacuum spectral density are $h_{\rm m}(c) = \sqrt{1+c^2}$ and $\rho(c) = 2c^2/\pi$. The initial coefficient $f(c)$ is real and nonnegative, with

$$\rho(c)|f(c)|^2 = \pi_0(c) = \frac{630}{c_1 - c_0} x^4 (1-x)^4 \mathbf{1}_{[c_0,c_1]}(c), \quad x = \frac{c - c_0}{c_1 - c_0}, \quad (c_0, c_1) = (0.4, 0.8). \tag{1}$$

This probability integrates to one. The zero extension of f is C^1, with quadratic endpoint zeros; it is not asserted to be C^∞. Use the bounded half-period creation blocks from our two-matter note, $\widetilde{A}_p = e^{\pi H_{\rm out}/2} A_p e^{-\pi H_{\rm in}/2}$, and $A_p(t) = e^{-iH_{\rm out}t} A_p e^{iH_{\rm in}t}$. The physical states are

$$v_s(t) = \widetilde{A}_q(t)\widetilde{A}_p(0) f, \qquad v_t(t) = \widetilde{A}_p(0)\widetilde{A}_q(t) f.$$

The distinct species make their action on the matter vacuum lie in the two-particle sector. Define

$$X(t) = \langle v_s(t), v_t(t)\rangle, \quad N_s = \|v_s\|^2, \quad N_t = \|v_t\|^2, \quad D(t) = N_s + N_t - 2\,\mathrm{Re}\, X(t) = \|v_s - v_t\|^2.$$

The norms are independent of t. This is a regulated two-creation order difference; $\widetilde{A}$ is not identified with a self-adjoint field. Retain

$$g(x) = \frac{x e^{\pi/(\sqrt{1+x^2}+x)}}{1 + e^{-2\pi x}}, \qquad \widehat{R}_p(a,b) = \frac{A - B}{8\pi(A+C)(B+C)},$$

where $(A, B, C) = (\cosh \pi(a+b), \cosh \pi(a-b), \cosh \pi p)$. Let $U_{00}(b, d; a, q, c, p)$ be the physical Racah matrix element fixed in our previous note from the LPRS kernel [1]. Then

$$X(t) = \int_{c_0}^{c_1} \mathrm{d}c\, \pi_0(c) \int_s^\infty \mathrm{d}a\ \mathrm{d}b\ \mathrm{d}d\, \mathcal{W}(a,b,d,c) U_{00}(b,d;a,q,c,p) e^{it[h_{\rm m}(a)+h_{\rm m}(c)-h_{\rm m}(b)-h_{\rm m}(d)]}, \tag{2}$$

$$\mathcal{W} = \frac{g(a)}{g(c)} \sqrt{\widehat{R}_q(a,b)\widehat{R}_p(b,c)\widehat{R}_p(a,d)\widehat{R}_q(d,c)}. \tag{3}$$

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There is no compact physical filter on b or d. Equation (2) is interpreted by the Hilbert-space limit of compact restrictions, without assuming absolute convergence of the fourfold integral. The positive norms are

$$N_s = \int \mathrm{d}c\, \pi_0(c) \int_s^\infty \mathrm{d}a\ \mathrm{d}b\, \frac{g(a)}{g(c)} \widehat{R}_q(a,b) \widehat{R}_p(b,c),$$

and N_t with p, q exchanged. The full Racah transform is unitary [2, 3]; its scalar compression is only a contraction. Consequently $|X| \le \sqrt{N_s N_t}$.

2 Analytic mass-tail control

Set $\delta(x) = (\sqrt{1+x^2} + x)^{-1}$ and

$$g_{\min} = \frac{c_0 e^{\pi\delta(c_1)}}{1 + e^{-2\pi c_0}}, \quad C_s = e^{\pi\delta(s)}, \quad C_0 = \frac{C_s - 1}{2\delta(s)}.$$

Then $g(c) \ge g_{\min}$ on the initial packet. Convexity of the exponential on $[0, \delta(s)]$ and $x\delta(x) \le 1/2$ give $g(x) \le x + C_0$. For $x \ge L$, also $g(x) \le x e^{\pi\delta(L)}$. The previously established bound is

$$0 < \widehat{R}_p(a,b) \le k_p(a-b), \qquad k_p(u) = \frac{1}{8\pi(\cosh \pi u + \cosh \pi p)} \le \frac{e^{-\pi|u|}}{4\pi}.$$

Let $v_{s,L}$ retain $a, b \le L$ and $v_{t,L}$ retain $a, d \le L$, where $L > c_1$. A union bound gives

$$\|v_s - v_{s,L}\|^2,\ \|v_t - v_{t,L}\|^2 \le T_a + T_b, \tag{4}$$

$$T_a = \frac{e^{\pi\delta(L)} e^{-\pi(L-c_1)}}{16\pi^3 g_{\min}} \left[L^2 + L(3/\pi - c_0) + 3/\pi^2 - c_0/\pi \right], \tag{5}$$

$$T_b = \frac{e^{-\pi(L-c_1)}}{8\pi^4 g_{\min}} (L + C_0 + 2/\pi). \tag{6}$$

For (5), extend the intermediate mass to the real line:

$$\int_{\mathbb{R}} e^{-\pi|a-b|-\pi|b-c|}\, \mathrm{d}b = e^{-\pi(a-c)}(a - c + 1/\pi), \qquad a > L > c.$$

Use c_0 in the polynomial and c_1 in the exponential. For (6), first bound

$$\int_s^\infty (a + C_0) e^{-\pi|a-b|}\, \mathrm{d}a \le 2(b + C_0)/\pi + 2/\pi^2.$$

The remaining integrals are elementary. Write $A_p = \int k_p = p/(4\pi \sinh \pi p)$ and $B = 1/(2\pi^3) \ge \int |u| k_p(u)\, \mathrm{d}u$. Extending both mass differences to the full plane gives

$$N_s, N_t \le \mathcal{N}_{\max} = \frac{(c_1 + C_0) A_p A_q + (A_p + A_q) B}{g_{\min}}. \tag{7}$$

Independently of the numerical norms,

$$|X - X_L| \le 2\sqrt{(T_a + T_b)\mathcal{N}_{\max}}. \tag{8}$$

Our parameters give $\mathcal{N}_{\max} \simeq 8.42931\, 10^{-4}$. At $L = 12$, the positive norm tail is at most $1.16353\, 10^{-16}$, and the crossed absolute tail is at most $6.26346\, 10^{-10}$. The bound for $D - D_L$ is $2(T_a + T_b) + 4\sqrt{(T_a + T_b)\mathcal{N}_{\max}}$. These control mass truncation only, not quadrature, special-function truncation, or floating-point error.

3 The exact spherical convolution

The actual local 000 coefficient simplifies to

$$E_{abp} = C_p \sqrt{\frac{\cosh \pi b}{\cosh \pi a}} H_p(a,b), \qquad C_p = \frac{1}{4\sqrt{\pi}} \sqrt{\frac{\sinh \pi p}{p}}, \tag{9}$$

where

$$H_p(a,b) = r_p(a-b) - r_p(a+b), \qquad r_p(x) = (\cosh \pi x + \cosh \pi p)^{-1}.$$

This follows by cancellation in the fixed BD/KG vertex, with no new insertion convention. Since r_p is even and integrable, H_p acts by convolution on an odd extension. Thus

$$\int_0^\infty H_q(a,b) H_p(b,c)\, \mathrm{d}b = \int_0^\infty H_p(a,b) H_q(b,c)\, \mathrm{d}b. \tag{10}$$

Each physical CG chain has $J_b(0000) = 1$. The four-external-spherical component of the zero-time order difference, before the common $f(c)$ and half-period factor, is

$$\Delta_{0000}(a,c) = -C_p C_q \sqrt{\frac{\cosh \pi c}{\cosh \pi a}} \int_0^s \mathrm{d}b\, [H_q(a,b) H_p(b,c) - H_p(a,b) H_q(b,c)]. \tag{11}$$

The regulator multiplies this by the positive real factor $e^{\pi[h_{\mathrm{m}}(a) - h_{\mathrm{m}}(c)]/2}$. For this component, the order difference is exactly the missing low-mass interval. Its disappearance at $s = 0$ does not prove that other spherical components or the full order difference vanish there.

4 A nonzero finite-norm witness

Put $P = \cosh \pi p$, $Q = \cosh \pi q$, $u = \cosh \pi a$, $v = \cosh \pi b$, and $D_P(u,v) = u^2 + v^2 - 1 + 2Puv + P^2$. Then $H_q / H_p = D_P / D_Q$ and

$$\frac{\partial}{\partial u} \frac{D_P}{D_Q} = \frac{2(Q-P)[vu^2 + (P+Q)u + v(1 + PQ - v^2)]}{D_Q^2}. \tag{12}$$

If $q > p$ and $PQ \geq \sinh^2 \pi s$, this is strictly positive for $0 < b < s$. For $a > c$, the integrand in (11) is strictly positive and $\Delta_{0000}(a,c) < 0$. Our sign-condition margin is about 28.1211.

Take a kinematic finite-K test state with all four spherical modes zero and $\chi(a,c) = \mathbf{1}_{[1.1,1.4]}(a) f(c)$. Its group average has finite physical squared norm

$$N_\chi = \int_{1.1}^{1.4} \mathrm{d}a \int_{0.4}^{0.8} \mathrm{d}c\, |f(c)|^2 K_4(a,q,c,p), \tag{13}$$

$$K_4(a,q,c,p) = \int_0^\infty \mathrm{d}r\, \sinh^2 r \prod_{n=a,q,c,p} \frac{\sin(nr)}{n \sinh r}. \tag{14}$$

Finiteness follows from the previous four-body group average for compact mass and finite spherical support. Its pairing with the actual regulated order difference is

$$Z_\chi = \int_{1.1}^{1.4} \mathrm{d}a \int_{0.4}^{0.8} \mathrm{d}c\, |f(c)|^2 e^{\pi[h_{\mathrm{m}}(a) - h_{\mathrm{m}}(c)]/2} \Delta_{0000}(a,c) < 0. \tag{15}$$

The physical Cauchy inequality proves

$$D(0) \geq |Z_\chi|^2 / N_\chi > 0. \tag{16}$$

This analytic witness does not depend on resolving a small difference of numerically integrated $6j$ overlaps. Quadrature gives

$$Z_\chi \simeq -1.1370592534\,10^{-5}, \quad N_\chi \simeq 0.69678827298, \quad |Z_\chi|^2/N_\chi \simeq 1.85552\,10^{-10}.$$

These constants are not interval-certified. The norm is also checked by an independent radial group integral. For fixed external labels,

$$H_p(a,b) = \frac{2\pi b \sinh \pi a}{(\cosh \pi a + P)^2} + O(b^3).$$

Hence (11) is $O(s^3)$, with its coefficient obtained by integrating the product of the linear terms. This is not a full $D \sim s^6$ theorem.

5 Accelerating the known conformal kernel

Each ${}_4F_3(1)$ in LPRS has parameter excess exactly one. Its nth term has the asymptotic form

$$t_n \sim C n^{-2} \exp\left(\sum_{k\geq 1} \ell_k n^{-k}\right), \qquad C = \frac{\prod_j \Gamma(b_j)}{\prod_i \Gamma(a_i)},$$

$$\ell_k = \frac{(-1)^{k+1}}{k(k+1)}\left[\sum_i B_{k+1}(a_i) - \sum_j B_{k+1}(b_j) - B_{k+1}(1)\right].$$

This is the standard Gamma-ratio expansion [4]; B_k are Bernoulli polynomials. With $c_0 = 1$, $c_k = k^{-1}\sum_{j=1}^{k} j\ell_j c_{k-j}$, sum $n < N$ explicitly and approximate the remaining tail by

$$C \sum_{k=0}^{K} c_k \zeta(k+2, N),$$

using the Hurwitz zeta series [5]. The code batches complex128 arithmetic and retains both Omega terms, both shadow terms, and the physical phases. The main setting is $(N, K) = (128, 12)$; refinement uses $(256, 16)$. There is no uniform complex-parameter remainder certificate. Forty-five-digit comparisons cover six influential packet nodes, one maximally ill-conditioned pilot node, and six seeded generic nodes. The largest influential-node relative error is about $1.96\,10^{-11}$. The ill-conditioned far node has about 1.97% relative error, retained in the record. Its small weight alone does not certify the integral. Condition-number times machine epsilon is only a sensitivity proxy.

6 Physical quadrature and remaining uncertainty

The final grid uses eight Gauss nodes per mass interval, 12 initial-mass nodes, and $L = 12$, for 2,107,392 complex kernel evaluations. Five preserved runs contain 3,442,832 kernel points in total. The positive norms, computed separately with 32 nodes per interval, 16 initial nodes, and $L = 20$, are

$$N_s \simeq 6.0532837979\,10^{-5}, \qquad N_t \simeq 6.0662442257\,10^{-5}.$$

The table reports approximate values, not certified error intervals.

t	$\mathrm{Re}\, X$	$\mathrm{Im}\, X$	$D(t)$	$\lvert X\rvert/\sqrt{N_s N_t}$
0	$6.02722782\,10^{-5}$	$-6.9\,10^{-19}$	$6.50724\,10^{-7}$	.994631
.5	$5.88726863\,10^{-5}$	$-5.61586481\,10^{-6}$	$3.44991\,10^{-6}$	.975945
1	$5.46710732\,10^{-5}$	$-1.11360698\,10^{-5}$	$1.18531\,10^{-5}$	.920725
2	$3.85245427\,10^{-5}$	$-1.98191785\,10^{-5}$	$4.41462\,10^{-5}$	.714940

The tiny imaginary entry at zero is retained as a floating-point residual. Changing from six to eight mass nodes at $L = 12$ (also changing the initial grid from 10 to 12 nodes and the series setting) changes the four complex overlaps by at most $1.02\,10^{-11}$. Changing the series setting on the identical pilot grid changes its integrated overlaps by at most $2.04\,10^{-20}$. The $L = 10$ to 12 six-node comparison, also changing the series setting, differs by at most $8.53\,10^{-15}$. These mixed refinements are stated explicitly; they are not independent one-variable error estimates.

Twenty-four vertex, cutoff-identity, positive-norm, tail-primitive, and witness checks pass. The independent emission/rigging replay of all five raw grids agrees with their stored overlap sums within $2\,10^{-17}$ absolute (the verifier's threshold). The table obeys the physical Cauchy bound and the independent lower bound (16). A nonzero value is proved by the witness; its full magnitude in the table remains a numerical estimate. The initial support is integrated in full. Other masses use piecewise Gauss–Legendre intervals with breaks at $s, 1, 2, 4, 6, 8, 10, 12$. Raw complex kernels, coordinates, weights, conditions, dependency versions, and hashes are retained. A separate replay reconstructs the weight from the original emission amplitudes and rigging weights.

Mass-tail bounds and the sign witness are analytic model statements. Grid changes, special-function comparisons, and whole-grid acceleration changes are numerical evidence, not a total error certificate. Complete Fock domains, identical-species Bose contributions, general preparations, finite gravity, and the full holographic dual remain open.

References

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[4] NIST DLMF, Section 5.11, especially (5.11.8).

[5] NIST DLMF, Section 25.11, especially (25.11.1).